

Software structure and ecosystem

The software is designed as a distributed system with multiple processors. This architecture ensures that the investments made in legacy software for the cellular system protocol stack processing can be safeguarded and incorporated in the new system solution. It also enables decoupling of the real-time critical tasks between the communication subsystems and the multimedia processor – so the signal processing tasks of the cellular subsystem don't compete for computing power with those of the audio/video processor. The result is a functional, extensible platform that can be implemented quickly with minimal capital outlay, freeing up resources to create added-value applications software and end-user services.

Type Number/Description	Package/Footprint
PNX4002 TV-out IC High performance image processing (embedded Image Signal Processor) up to 3.1 megapixels (RGB,YUV,JPEG formats), Image and color enhancement, support for TV Out (CVBS NTSC/PAL/SECAM), analog output for viewing still pictures and video on a TV set, graphics functions, overlay of several visual objects (text, icons, pictures) in real-time, dual display support (up to QVGA), 18-bit color depth, RAM-based or RAM-less LCD.	LFBGA240 10x10 mm
LP8773 QCIF+ color display driver Supports AMLCD display up to 1.9 inches in diameter with up to 262k colors and up to QCIF+ resolution (176 x 220 pixels). Overall display dimensions (w x h x d): 29.8 x 37.3 x 2.6 mm. Pixel dimensions (w x h): 56.5 (x 3) x 169.5 _m.	N/A
TEA5767 Low-power FM radio Electronically tuned FM stereo radio with fully integrated IF selectivity and demodulation. Completely adjustment-free and only requires a minimum of small and low-cost external components. For use with European, US, and Japanese FM bands.	HVQFN40 6x6 mm
BGW200 Low-power 802.11b WLAN SiP Complete solution combines a baseband/MAC, RF transceiver, and power amplifiers. Delivers industry-leading standby power consumption, superior “no host load” architecture, latest QoS and security standards, and coexists with Bluetooth 1.2.	HVQFN68 10x15 mm
PN531 NFC controller Near Field Communication (NFC) controller with 80C51-based microcontroller, 32k of ROM, and 1k of RAM. Operates off a 2.5-to-3.6V power supply or a USB connection, with typical operating voltage of 3.3V.	HVQFN40 6x6 mm

Linux Open OS and ecosystem

To make it easier to integrate various middleware and application software components, the application engine uses an embedded version of the Linux Open operating system. Using Linux OS lets system integrators avoid having to integrate a legacy cellular protocol processor stack or its associated proprietary operating system. It also lets integrators take advantage of the many skilled resources and application software components available on the open market.

Nexperia Mobile Developer's Kit

The Nexperia Mobile Developer's Kit has all the elements needed to start working with a Nexperia Cellular System Solution platform. This simple-to-run yet powerful kit, developed in collaboration with ARM and Metrowerks, includes industry-leading ARM tools and uses the popular Metrowerks Codewarrior for its development environment. The Kit is shipped ready to run on a standard Windows PC. As an added service, dedicated local support and customer training are also available.

The System Solution 9100 Developer's Kit comes with a Linux Integrated Development Environment (IDE) based on the MontaVista DevRocket™ IDE. The Kit also integrates an application framework based on Trolltech Qtopia™ development tools.



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Nexperia™ System Solution 9100

Multimedia-enabled system solution for GSM/GPRS/EDGE/UMTS smartphones running Linux Open OS

This dual-chip, ARM926-based GSM/GPRS/EDGE/UMTS platform, designed for the high-end smartphone market, offers an unprecedented mix of multimedia features, plus industry-leading power consumption and an open OS.

Key features

- Complete, market-proven design for next-generation GSM/GPRS/EDGE/UMTS multimedia handsets
 - Mature, reliable hardware/software solution
 - Fully validated Interoperability Testing (IOT)
 - More than 150 million Nexperia System Solutions already deployed
- Optimized core functionality based on Nexperia Cellular System Solution 6120
 - PCF5213 Nexperia multimedia baseband with 104-MHz ARM926 core (Linux OS)
 - UAA3587 GSM/GPRS/EDGE RF SiP
 - BGY286 PA module
 - PCF50603 PMU and UBA2008 charge switch
 - Protocol stack software
 - Optimized Multimedia functionality:
 - PNX4008 Nexperia multimedia application engine with Open OS Linux and 208-MHz ARM926 core
 - BGB204 Bluetooth® module
 - ARM926-based application processing, WLAN, NFC, TV-on-Mobile (DVB-H), TV Out, and more
 - Easy customization with added multimedia options
- Supported by Nexperia Mobile Developer's Kit
- Dedicated local support and customer training are available

Designed for the smartphone and feature-rich mobile handset market, the Nexperia System Solution 9100, an ARM926-based GSM/GPRS/EDGE/UMTS multimedia platform, provides next-generation multimedia features in a flexible format that is easy to differentiate. It offers guaranteed telecom performance, integrated messaging and connectivity, plus advanced video, imaging, and audio functions.

The platform builds on a core chipset that minimizes power consumption to deliver industry-leading standby and talk times. Manufactured in high volume, it comes fully GCF/PTCRB tested, so OEMs and ODMs can save time while delivering mature, reliable technology.

For further time-to-market savings, the System Solution 9100 uses an open, Linux-based operating system for the ARM926 processor and its associated multimedia application processor. This makes it easier to integrate various middleware and application software components and expands the options for multimedia add-ons.

Market-proven solutions

Each Nexperia cellular system solution is a carefully designed combination of powerful core hardware and proven, mature software. More than 150 million Nexperia-based systems have been brought to market and more than one in every ten GSM/GPRS handsets uses a Nexperia solution, which makes these platforms some of the most extensively proven designs available.

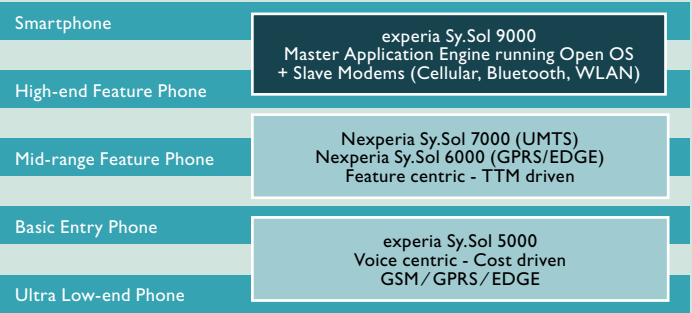




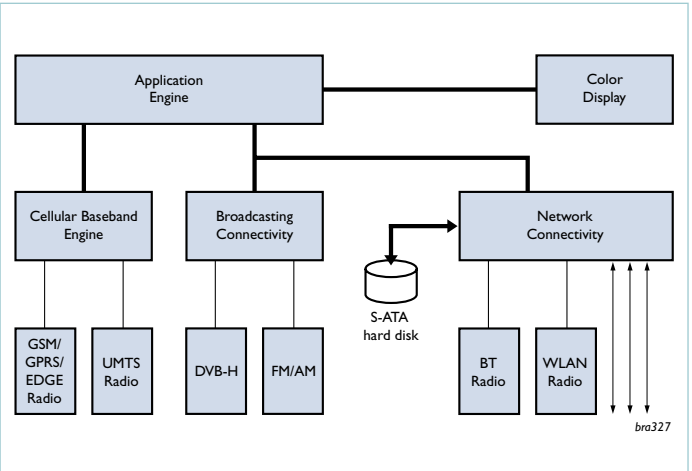
Optimized core chipset

The core chipset includes a Nexperia baseband for multi-band operation, a Nexperia application engine for multimedia applications, an RF transceiver, a power amplifier, a power management unit, a battery charger, and a complete protocol stack. The result is an optimized design that provides outstanding standby and connection times.

Type Number/Description	Package/Footprint
PCF5213 Nexperia multimedia baseband Quad-band (850/900/1800/1900) baseband with Class 12 EGPRS capability.Architecture combines ARM9 CPU with 16-bit DSP to deliver usable rates of up to 300 kbits. High integration means baseband runs advanced applications directly, without external circuitry.	LF8GA228 12x12 mm or LF8GA240 10x10 mm
PNX4008 Nexperia multimedia application engine 208-MHz ARM926-EJ processor with dedicated 32k instruction and 32k data caches, supported by dedicated subsystems for video, audio, graphics, and security. Offers high-resolution imaging, video record/playback, video telephony, enhanced audio/music, 3D gaming, and security functions.	LF8GA320 13x13 mm
UAA3587 GSM/GPRS/EDGE RF SiP Highly integrated, small-footprint RF transceiver with integrated Tx baluns, loop filters, decoupling capacitors, and fractional-N RF synthesizer with AFC control. Semi-integrated 26-MHz oscillator with fully integrated supply regulator.	HVQFN40 6x6 mm
BGY284 PA module Power amplifier with integrated power control loop. High-efficiency operation (>50% high bands, > 55% low bands). Fixed-gain EDGE amplifier. Integrated 50_ input and output impedance matches.	Module 8x10 mm
PCF50603 power management unit (PMU) Complete, highly integrated PMU with sophisticated on/off control, programmable power supplies, flexible battery management, a SIM-card interface, and support for audio peripherals.	HVQFN48 6x6 mm
UBA2008 charge switch Intelligent charge switch for pulse-mode charging applications. Low-ohmic fast charge switch (charger-to-battery), low-ohmic reverse mode switch (battery-to-charger), and constant current source (charger-to-battery) for slow or top-off charging.	HVSON10 3x3 mm
BGB204 Bluetooth SiP Complete solution combines a Bluetooth 1.2 baseband with embedded Flash and a full radio. Only an external clock source, from the System Solution 9100 cellular platform, and an antenna are required for full operation. Coexists with 802.11 WLAN.	HVQFN 7x7 mm



The Philips family of Nexperia cellular System Solutions



Nexperia cellular System Solution 9100 application block diagram

Multimedia application engine

The PNX4008 Nexperia multimedia application engine acts as the central media processing unit, implementing the system “switchboard” and managing all the incoming and outgoing media streams. It runs the Linux Open operating system. Time-critical operations can be spread between the baseband and the application engine, reducing the overhead of multimedia functions. The application engine connects directly to the video peripherals and the cellular baseband, and has a link to the devices for connectivity and broadcasting functions. This makes it possible to add and remove communication pipes for a more flexible, customized implementation.

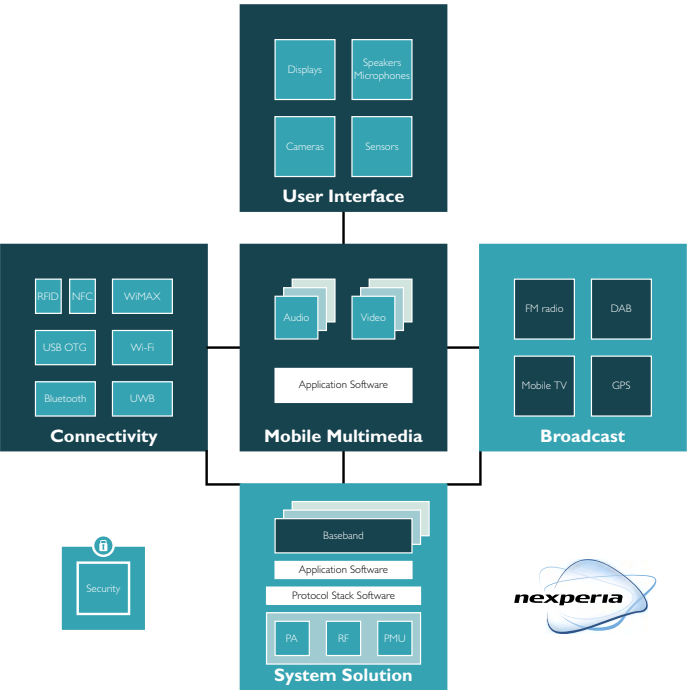
Built-in multimedia features

The System Solution 9100 provides an unprecedented mix of multimedia features that raise the core chipset to a higher level of multimedia performance.

Broadcast capabilities go beyond the standard AM/FM receiver and MP3 capability to include capabilities for TV (DVB-H) reception. Recovered OFDM-coded signals are forwarded to the application engine for stream decoding, post-processing, and display. The System Solution 9100 also supports TV Out, for interaction with a home multimedia system.

Connectivity technologies are managed by a subsystem that governs operation of Near Field Communication (NFC), Bluetooth, WLAN, USB 2.0, Ethernet, and interfaces to local mass-storage devices. A fast serial link connects the subsystem to the application engine (a dedicated processor can be provided for the protocol stacks associated with the connectivity subsystem).

Further options for multimedia, broadcast, and connectivity
Built around a highly flexible platform-based architecture, the System Solution 9100 can quickly be extended with additional features for completely tailored performance. In particular, unique Philips LifeVibes™ algorithms provide enhanced performance in video, still imaging, audio, voice, and security applications. Design teams can also add their own software or software supplied by a third party to the basic system.



Nexperia cellular System Solution 9100 core software

Telecom Features

- Quad band GPRS/EDGE Class 10
- Release 99
- SMS, EMS rel. 4
- WAP 2.0 and MMS 2.0
- Audio codecs: HR, FR, EFR, AMR
- Open audio interface
- Pulse mode charging (with UBA2008)
- Hands Free
- TTY
- IrDA and MMC enabled (drivers)
- USB boot, enabling USB only bottom connector
- Mono Interference Cancellation (MIC)
- Multi TA / 07.10 allows modem application in with multi-tasking OS
- Hardware security
- A-GPS, access to assistance data

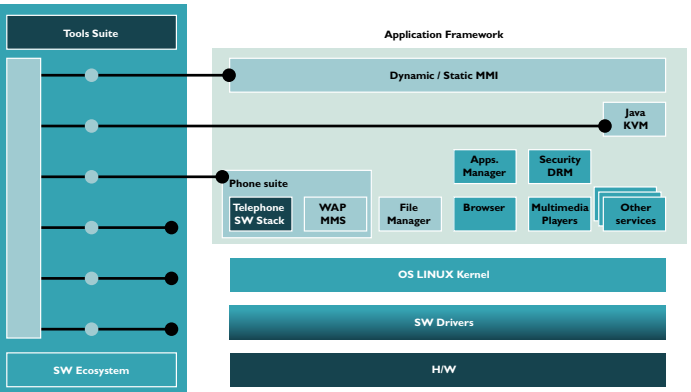
Multimedia Features

- Support Linux Kernel CEE3.1 From Montavista Inc.
- Camera support resolution up to 4 Megapixel receiving inputs in YUV (4:2:2) format
- Video:
 - LifeVibes Multimedia Video Encoder Service Video (VES) Video Player Service (VPS)
 - Record/Playback MPEG-4 SP, H.263 Real-time encoding decoding and duplex CIF resolution @ 30 fps
 - Video telephony – Full duplex, low-latency video encode/decode mode
 - Divx5
 - Post/pre-processing (face tracking, stabilization...)
 - Hardware Assisted Video Engine
 - A/V progressive download/ streaming
 - 3D Gaming: Open GL-ES compliant /3D graphics accelerator
- Image:
 - Still image capture up to 4 Megapixels
 - Still picture viewer – JPEG baseline, GIF87, animated GIF, PNG, WBMP
 - Pre-post processing (Smart imaging)

- Audio:
 - LifeVibes Voice
 - Dedicated DSP for all audio activities
 - MP3,AAC, WB-AMR
 - AAC+, AAC+ enhanced
 - Real Audio V8/V9
 - Stereo encoding
 - Software polyphony16 tones polyphony at 8-kHz sampling rate
 - Voice Clarity audio enhancement
 - Extended Music format XMF and MFI
 - Handsfree
- J2ME JAVA TWI/JSR185
- File system: TFFS and FAT16
- Security:
 - LifeVibes TRUST DRM 2.0 agent (FL, CD, SD/XML, SD/WBXML)
 - HW/SW Security PKI, Symmetric and Hash Coprocessors
 - Random Number Generator with fuses for key generation

Connectivity Features

- USB 2.0 full-speed device (12 Mbps)
- RS232 (I)
- Bluetooth 1.2
- DVB-T/H
- TV OUT
- WLAN 802.11b/g/h
- NFC



Reference Architecture SW Ecosystem on Linux